

Module 06: Learning — The Little Robot's Mistake

Learning Objectives

1. Understand what **Learning** is (updating your brain's map).
2. Learn that mistakes and **Surprise** help us learn.
3. See how practicing makes us better predictors.

Introduction: The Little Robot's Mistake

Beep-Boop the Robot loved to play fetch. He had a map in his computer brain. **Map Rule #1:** "Balls are light. I can throw them far!"

One day, Beep-Boop saw a bowling ball. It looked like a ball. "I predict this will be light!" said Beep-Boop. He took **Action**. He tried to throw it. *CLUNK!*

It was HEAVY! Beep-Boop fell over. *Crash!* He was **Surprised**. His prediction was wrong!

"Oh no!" beeped the robot. "My map was wrong." He took out a pencil and fixed his map. **New Map Rule:** "Some balls are heavy. Check first!"

Beep-Boop **Learned** something new because he made a mistake.

Key Concepts

1. What is Learning?

Learning is fixing the map in your head.

- When you are a baby, you don't know that stoves are hot.
- If you touch one... *Ouch!* You are surprised.
- You learn: "Stoves are hot!" You update your map.

2. Surprise is Good

In school, sometimes it feels bad to be wrong. But for your brain, being wrong is the *only* way to learn!

- If you guess everything right, your map is already perfect.
- If you are **Surprised**, it means you have something new to learn.
- **Surprise** is the teacher!

3. Practice Makes Perfect

Why do we practice sports or reading? To make our brain's predictions better!

- First try: "I predict I will hit the ball." (Miss! Surprise!) -> Fix Map.
- Second try: "I predict I will hit the ball." (Hit!) -> Good Map.

Activities

Activity 1: The Heavy Box Trick

Get two identical boxes. Put a brick in one and a feather in the other. Ask a friend to lift them. Watch their face when they lift the heavy one! That look is **Surprise**. They just learned!

Activity 2: Drawing from Memory

Look at a toy for 10 seconds. Hide the toy. Draw it. Look at the toy again. Did you miss anything? Fix your drawing. That is **Learning**!

Summary

Learning happens when our predictions are wrong. When we are **Surprised**, we fix the maps in our heads so we can guess better next time. Mistakes are just invitations to learn!

References

- *Beautiful Oops!* by Barney Saltzberg (A book about how mistakes can be beautiful.)

- *The Girl Who Never Made Mistakes* by Mark Pett and Gary Rubinstein